Worksheet 6.4 — Applications: Area Between Curves and Kinematics

Mathematics: Analysis & Approaches SL (IB) · Unit 6: Integral Calculus

The Big Ideas

Area between curves and total-distance kinematics both split at crossing points and use absolute values.

- Area $= \int_a^b [f(x) - g(x)]\,dx$, with a, b the intersection points.
- Total distance $= \int |v(t)|\,dt$ — split at every zero of $v(t)$.

Worked Examples

Example 1 — Area between curves

Find the area enclosed between $f(x) = x + 6$ and $g(x) = x^2$.

Solution. Intersect: $x + 6 = x^2 \Rightarrow x^2 - x - 6 = 0 \Rightarrow (x-3)(x+2) = 0$: $x = -2, 3$. f on top: $\int_{-2}^{3}[(x+6) - x^2]dx = \left[\frac{x^2}{2} + 6x - \frac{x^3}{3}\right]_{-2}^{3} = (4.5 + 18 - 9) - (2 - 12 + \frac{8}{3})$.

Conclusion: $13.5 - (-7.33) = \dfrac{125}{6}$.

Example 2 — Displacement via integration

$v(t) = 6t - 18$ m/s. Find displacement from $t = 0$ to $t = 5$.

Solution. $[3t^2 - 18t]_0^5 = (75 - 90) - 0$.

Conclusion: -15 m.

Example 3 — Total distance via integration

Using the same $v(t) = 6t - 18$, find total distance from $t = 0$ to $t = 5$.

Solution. $v = 0$ at $t = 3$. $\int_0^3 (6t-18)dt = (27 - 54) - 0 = -27$. $\int_3^5 (6t-18)dt = (75 - 90) - (27 - 54) = -15 - (-27) = 12$.

Conclusion: total $= 27 + 12 = 39$ m.

Example 4 — Comparing accumulated rates

Two profit rates (\$1000s/month) are $p_1(t) = t + 4$, $p_2(t) = 0.4t^2$. Find where $p_1 > p_2$ and the extra profit over that interval.

Solution. $t + 4 = 0.4t^2 \Rightarrow 0.4t^2 - t - 4 = 0 \Rightarrow t^2 - 2.5t - 10 = 0 \Rightarrow t = \dfrac{2.5 \pm \sqrt{6.25 + 40}}{2} \approx 4.55$ (positive root). Extra: $\int_0^{4.55}[(t+4) - 0.4t^2]dt \approx \left[\frac{t^2}{2} + 4t - \frac{0.4t^3}{3}\right]_0^{4.55} \approx 10.35 + 18.2 - 12.58$.

Conclusion: Company 1 leads for ≈ 4.55 months, gaining $\approx$ \$15,970.

Example 5 — Area between a line and a cubic-adjacent pair (via a shared factor)

Find the area enclosed between $f(x) = 4x$ and $g(x) = x^3$ for $x \geq 0$ (find all intersection points first).

Solution. $4x = x^3 \Rightarrow x^3 - 4x = 0 \Rightarrow x(x^2 - 4) = 0 \Rightarrow x = 0, \pm 2$; restricting to $x \geq 0$: $x = 0, 2$. On $(0, 2)$, $f > g$ (check $x = 1$: $4 > 1$).

Area $= \int_0^2 (4x - x^3)dx = \left[2x^2 - \frac{x^4}{4}\right]_0^2 = 8 - 4$.

Conclusion: area $= 4$ square units.

Example 6 — Displacement and distance with a cubic velocity

$v(t) = t^2 - 4$ m/s. Find the displacement and total distance from $t = 0$ to $t = 3$.

Solution. Displacement: $\int_0^3 (t^2 - 4)dt = \left[\frac{t^3}{3} - 4t\right]_0^3 = 9 - 12 = -3$ m.

$v = 0$ at $t = 2$: $\int_0^2 (t^2 - 4)dt = \frac{8}{3} - 8 = -\frac{16}{3}$; $\int_2^3 (t^2 - 4)dt = (9 - 12) - (\frac{8}{3} - 8) = -3 + \frac{16}{3} = \frac{7}{3}$.

Conclusion: displacement $= -3$ m; total distance $= \frac{16}{3} + \frac{7}{3} = \frac{23}{3} \approx 7.67$ m.

Example 7 — Area between curves with a shared tangent point

Find the area enclosed between $f(x) = x^2$ and $g(x) = 2x - 1$ (note: these touch rather than cross).

Solution. $x^2 = 2x - 1 \Rightarrow x^2 - 2x + 1 = 0 \Rightarrow (x - 1)^2 = 0 \Rightarrow x = 1$ (a repeated root — the curves are tangent, not crossing).

Conclusion: since there's only one intersection (a tangency), the curves don't enclose any region — area $= 0$.

Example 8 — Application — net work from a variable force

A variable force is $F(x) = 10 - 2x$ N, applied over $x \in [0, 4]$ m. Find the net work done (force integrated over distance).

Solution. $W = \int_0^4 (10 - 2x)dx = [10x - x^2]_0^4 = 40 - 16$.

Conclusion: $W = 24$ J.

Example 9 — Total distance for a velocity with three sign changes

$v(t) = (t - 1)(t - 3)(t - 5)$ for $0 \leq t \leq 6$. Identify every time the particle is at rest, without fully computing the total distance.

Solution. $v(t) = 0$ when any factor is zero.

Conclusion: $t = 1, 3, 5$ — three rest points, meaning the total-distance calculation would need four separate integrals (over $[0, 1], [1, 3], [3, 5], [5, 6]$).

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Find the area enclosed between $f(x) = 12 - x^2$ and $g(x) = x^2 - 4$.

Question 2

Find the area enclosed between $f(x) = x + 3$ and $g(x) = x^2 - 3$.

Question 3

$v(t) = 4t - 16$ m/s. Find the displacement from $t = 0$ to $t = 6$.

Question 4

Using the same $v(t)$, find the total distance travelled from $t = 0$ to $t = 6$.

Question 5 — Challenge

A particle's velocity is $v(t) = 3t^2 - 18t + 24$ m/s; find the total distance travelled from $t = 0$ to $t = 5$ (first find when it's at rest).

Question 6 — Challenge

Find the area enclosed between $f(x) = x^2 - 4x$ and $g(x) = -x^2 + 4x + 8$.

Question 7

Find the area enclosed between $f(x) = 9x$ and $g(x) = x^3$ for $x \geq 0$ (find all intersection points first).

Question 8 — Challenge

$v(t) = t^2 - 9$ m/s. Find the displacement and total distance from $t = 0$ to $t = 4$.

Question 9

Determine the area enclosed between $f(x) = x^2 - 2x + 1$ and $g(x) = 2x - 3$ (check whether these curves cross or merely touch first).

Question 10

A variable force is $F(x) = 20 - 4x$ N applied over $x \in [0, 5]$ m; find the net work done.

Question 11

$v(t) = (t-2)(t-4)(t-6)$ for $0 \le t \le 7$; identify every time the particle is at rest.

Question 12 — Challenge

Find the area enclosed between $f(x) = x^2 - 6x + 9$ and $g(x) = 9 - x^2$.

Question 13 — Challenge

Two cost rates (\$100s/week) are $c_1(t) = 20 - t$, $c_2(t) = 0.5t^2$; find where $c_1 > c_2$ and the total savings from using process 1 over that interval.

Question Recap & Answer Key

Question Recap

1. Find the area enclosed between $f(x) = 12 - x^2$ and $g(x) = x^2 - 4$.
2. Find the area enclosed between $f(x) = x + 3$ and $g(x) = x^2 - 3$.
3. $v(t) = 4t - 16$ m/s. Find the displacement from $t = 0$ to $t = 6$.
4. Using the same $v(t)$, find the total distance travelled from $t = 0$ to $t = 6$.
5. A particle's velocity is $v(t) = 3t^2 - 18t + 24$ m/s; find the total distance travelled from $t = 0$ to $t = 5$ (first find when it's at rest).
6. Find the area enclosed between $f(x) = x^2 - 4x$ and $g(x) = -x^2 + 4x + 8$.
7. Find the area enclosed between $f(x) = 9x$ and $g(x) = x^3$ for $x \geq 0$ (find all intersection points first).
8. $v(t) = t^2 - 9$ m/s. Find the displacement and total distance from $t = 0$ to $t = 4$.
9. Determine the area enclosed between $f(x) = x^2 - 2x + 1$ and $g(x) = 2x - 3$ (check whether these curves cross or merely touch first).
10. A variable force is $F(x) = 20 - 4x$ N applied over $x \in [0, 5]$ m; find the net work done.
11. $v(t) = (t - 2)(t - 4)(t - 6)$ for $0 \leq t \leq 7$; identify every time the particle is at rest.
12. Find the area enclosed between $f(x) = x^2 - 6x + 9$ and $g(x) = 9 - x^2$.
13. Two cost rates (\$100s/week) are $c_1(t) = 20 - t$, $c_2(t) = 0.5t^2$; find where $c_1 > c_2$ and the total savings from using process 1 over that interval.

✓ Answer Key

1. $12 - x^2 = x^2 - 4 \Rightarrow 16 = 2x^2 \Rightarrow x = \pm 2\sqrt{2}$; area $= \int_{-2\sqrt{2}}^{2\sqrt{2}} (16 - 2x^2)dx = \left[16x - \frac{2x^3}{3}\right]$, evaluating gives $\frac{128\sqrt{2}}{3} \approx 60.34$
2. $x + 3 = x^2 - 3 \Rightarrow x^2 - x - 6 = 0 \Rightarrow (x - 3)(x + 2) = 0$: $x = -2, 3$; area $= \int_{-2}^{3} [(x + 3) - (x^2 - 3)]dx = \int_{-2}^{3} (6 + x - x^2)dx = \frac{125}{6}$
3. $[2t^2 - 16t]_0^6 = 72 - 96 = -24$ m
4. $v = 0$ at $t = 4$; $\int_0^4 (4t - 16)dt = (32 - 64) - 0 = -32$; $\int_4^6 (4t - 16)dt = (72 - 96) - (32 - 64) = -24 - (-32) = 8$; total $= 32 + 8 = 40$ m
5. $v = 0$ at $t = 2, 4$; $\int_0^2 (3t^2 - 18t + 24)dt = (8 - 36 + 48) - 0 = 20$; $\int_2^4 (\cdot)dt = (64 - 144 + 96) - 20 = 16 - 20 = -4$; $\int_4^5 (\cdot)dt = (125 - 225 + 120) - 16 = 20 - 16 = 4$; total $= 20 + 4 + 4 = 28$ m
6. $x^2 - 4x = -x^2 + 4x + 8 \Rightarrow 2x^2 - 8x - 8 = 0 \Rightarrow x^2 - 4x - 4 = 0 \Rightarrow x = 2 \pm 2\sqrt{2}$; area $= \int [(-x^2 + 4x + 8) - (x^2 - 4x)]dx = \int (-2x^2 + 8x + 8)dx$ over that interval ≈ 45.25
7. $9x = x^3 \Rightarrow x(x^2 - 9) = 0 \Rightarrow x = 0, \pm 3$; restrict $x \geq 0$: $x = 0, 3$; area $= \int_0^3 (9x - x^3)dx = \left[\frac{9x^2}{2} - \frac{x^4}{4}\right]_0^3 = 40.5 - 20.25 = 20.25$
8. displacement $= \int_0^4 (t^2 - 9)dt = [\frac{t^3}{3} - 9t]_0^4 = \frac{64}{3} - 36 = -\frac{44}{3}$; $v = 0$ at $t = 3$: $\int_0^3 (t^2 - 9)dt = 9 - 27 = -18$; $\int_3^4 (t^2 - 9)dt = (\frac{64}{3} - 36) - (9 - 27) = -\frac{44}{3} + 18 = \frac{10}{3}$; total distance $= 18 + \frac{10}{3} = \frac{64}{3} \approx 21.33$ m
9. $x^2 - 2x + 1 = 2x - 3 \Rightarrow x^2 - 4x + 4 = 0 \Rightarrow (x - 2)^2 = 0$: repeated root at $x = 2$, so the curves are tangent, not crossing — area $= 0$
10. $W = \int_0^5 (20 - 4x)dx = [20x - 2x^2]_0^5 = 100 - 50 = 50$ J
11. $t = 2, 4, 6$
12. $x^2 - 6x + 9 = 9 - x^2 \Rightarrow 2x^2 - 6x = 0 \Rightarrow 2x(x - 3) = 0 \Rightarrow x = 0, 3$; on $(0, 3)$, $g > f$ (check $x = 1$: $g = 8 > f = 4$); area $= \int_0^3 [(9 - x^2) - (x^2 - 6x + 9)]dx = \int_0^3 (6x - 2x^2)dx = [3x^2 - \frac{2x^3}{3}]_0^3 = 27 - 18 = 9$
13. $20 - t = 0.5t^2 \Rightarrow 0.5t^2 + t - 20 = 0 \Rightarrow t^2 + 2t - 40 = 0 \Rightarrow t = \frac{-2 + \sqrt{164}}{2} \approx 5.4$ (positive root); savings $= \int_0^{5.4} [(20 - t) - 0.5t^2]dt \approx [20t - \frac{t^2}{2} - \frac{t^3}{6}]_0^{5.4} \approx 108 - 14.58 - 26.24 \approx 67.2$ (\$100s), i.e. about \$6720